Q1: $AB = \begin{bmatrix} 3(1)+4(0) & 3(2)+4(-1) \\ 2(1)+1(0) & 2(2)+2(-1) \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$ $BA = \begin{bmatrix} 1(3)+2(2) & 1(4)+2(1) \\ 0(3)-1(2) & 0(4)-1(1) \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$

$(A+B)^T = \begin{bmatrix} 4 & 4 \\ 2 & 0 \end{bmatrix}^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$ $AB^T = \begin{bmatrix} 3(1)+4(2) & 3(0)+4(-1) \\ 2(1)+1(2) & 2(0)+1(-1) \end{bmatrix} = \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}$

trace(A) = 3+1 = 4 trace(AB) = 3+3 = 6

Q2: $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

find eigenvalues, eigenvectors

verify by showing $Mv = \lambda v$

$\det(M-\lambda I) = 0$

$\det\begin{pmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - (-1)^2 = 0$

$\lambda^2 - 4\lambda + 3 = 0$, $(\lambda-3)(\lambda-1) = 0$

$\lambda_1 = 3$

$(M-3I)v = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$-v_1 - v_2 = 0$ → $-x_1 = x_2$

$-x_1 - x_2 = 0$ $x_1 = 1$

$x_2 = -1$

$v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\lambda_2 = 1$

$(M-I)v = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$x_1 - x_2 = 0$ → $x_1 = x_2$

$-x_1 + x_2 = 0$

$v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$Mv_1 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$ ✓

$\lambda_1 v_1 = 3\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$

$Mv_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\lambda_2 v_2 = 1\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ ✓

verify

Q3: prove trace(AB) = trace(BA)

trace requires only diagonal of AB or BA

so instead of $(AB)_{ij} = \sum_{k=1}^{n} A_{ik}B_{kj}$

we only need $(AB)_{ii} = \sum_{k=1}^{n} A_{ik}B_{ki}$

so $\text{trace}(AB) = \sum_{i=1}^{m}(AB)_{ii} = \sum_{i=1}^{m}\sum_{k=1}^{n} A_{ik}B_{ki}$

because A_{ik} and B_{ki} are numbers not matrices,

$A_{ik}B_{ki} = B_{ki}A_{ik}$ so

$\text{trace}(AB) = \sum_{i=1}^{m}\sum_{k=1}^{n} A_{ik}B_{ki} = \sum_{i=1}^{n}\sum_{k=1}^{m} B_{ki}A_{ik} = \text{trace}(BA)$

hence $\text{trace}(AB) = \text{trace}(BA)$

Q4: inverse of $C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

$\det(C) = 4(6) - 2(7) = 16 \neq 0$

thus C^{-1} exists

$C^{-1} = \frac{1}{\det}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{10}\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$

$= \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}$

$C \cdot C^{-1} = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}\begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$= I$ ✓

Q_1: $P(X=k) = c \cdot \frac{3^k}{k!}$, $k=0,1,2\ldots$

a) normalization constant c

sum all probabilities $= 1 = \sum_{k=0}^{\infty} P(X=k)$

so $\sum_{k=0}^{\infty} c \cdot \frac{\lambda^k}{k!} = 1$ $k=\text{rate}=3$

so $c \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = 1$, $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}$

so $ce^{\lambda} = 1 \rightarrow c = \frac{1}{e^{\lambda}} = e^{-\lambda}$

When rate $=3$, $c = e^{-3} \simeq 0.05$

b) with $c = e^{-\lambda}$, $P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$

$E[X] = \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!} = \lambda e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$

$= \lambda e^{-\lambda}(e^{\lambda}) = \lambda$, for rate $=3$, $\lambda=3$

$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^k e^{-\lambda}}{k!} = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2$

$\text{Var}(X) = E[X(X-1)] + E[X] - (E[X])^2$

$= \lambda^2 + \lambda - (\lambda)^2 = \lambda$, for rate $=3$, $\lambda=3$

Q_2: $P(A)=0.5$, $P(B)=0.3$, $P(A|B)=0.6$

$P(A \cap B) = P(A|B) \cdot P(B) = 0.6(0.3) = 0.18$

$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.3} = 0.36$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.3 - 0.18 = 0.62$

Q_3: 3 coin flips H, H, T; H, T

a) $L(\theta) = P(Z|\theta) = \theta^k (1-\theta)^{n-k} = \theta^3(1-\theta)^2$

maximize $\frac{\partial P(Z|\theta)}{\partial \theta} = \frac{\partial}{\partial \theta}\left[\theta^3(1-\theta)^2\right] = 0$

$3\theta^2(1-\theta)^2 - 2\theta^3(1-\theta) = 0$

$\theta^2(1-\theta)\left[3(1-\theta) - 2\theta\right] = 0$

$3 - 3\theta - 2\theta = 0 \rightarrow 5\theta = 3$, $\theta = \frac{3}{5}$, so MLE $= 0.6$

b) $P(\theta) = \begin{cases} 2\theta, & 0 \le \theta \le 1 \\ 0 & \text{otherwise} \end{cases}$

$\text{argmax}_\theta \frac{P(Z|\theta)P(\theta)}{P(Z)} = \text{argmax}_\theta \left[P(Z|\theta)P(\theta)\right]$

$P(Z) = \int_0^1 P(Z|\theta)P\theta\, d\theta = \int_0^1 2\theta^4(1-\theta)^4 d\theta$

$= 2\int_0^1 (\theta^4 - 2\theta^5 + \theta^6)\, d\theta = 2\left[\frac{\theta^5}{5} - \frac{2\theta^6}{6} + \frac{\theta^7}{7}\right]_0^1 = \frac{2}{105}$

$P(\theta|Z) = \frac{2\theta^4(1-\theta)^2}{2/105} = 105\theta^4(1-\theta)^2$

$\theta_{MAP} = \text{argmax}_\theta \left[105\theta^4(1-\theta)^2\right]$

$\frac{d}{d\theta}\left[105\theta^4(1-\theta)^2\right] = 105\left[4\theta^3(1-\theta)^2 - 2\theta^4(1-\theta)\right] = 0$

$4(1-\theta) - 2\theta = 0 \rightarrow 6\theta = 4 \rightarrow \theta = 2/3$

Q_4: $x = 1, 3, 5, 7$

$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1+3+5+7}{4} = 4$

$E[x^2] = \frac{1}{n}\sum_{i=1}^{n} x_i^2 = \frac{1+3^2+5^2+7^2}{4} = 21$

$\sigma^2 = E[x^2] - \mu^2 = 21 - 4^2 = 5$

Q_5: $\sigma^2 = 4$ $\mu \sim N(0,1)$ $x = 5$

Likelihood: $P(x|\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

prior: $P(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(\mu-\mu_0)^2}{2\sigma_0^2}\right)$

$\mu_{MAP} = \text{argmax}_\mu P(\mu|x) = \text{argmax}_\mu \left[P(x|\mu)P(\mu)\right]$

$\ln P(\mu|x) = -\frac{(x-\mu)^2}{2\sigma^2} - \frac{(\mu-\mu_0)^2}{2\sigma_0^2} + c$

$\frac{d}{d\mu} \ln P(\mu|x) = \frac{x-\mu}{\sigma^2} - \frac{\mu-\mu_0}{\sigma_0^2} = 0$

so $\mu_{MAP} = \frac{\sigma_0^2 x + \sigma^2 \mu_0}{\sigma^2 + \sigma_0^2} = \frac{1(5) + 4(0)}{4+1} = \frac{5}{5} = 1$

Q_6: Prove $\text{Var}(x) = E(x^2) - (E(x))^2$

we know $\text{Var}(x) = E\left[(x - E[x])^2\right]$

$= E\left[x^2 - 2xE[x] + (E[x])^2\right]$

$= E[x^2] - E[2xE[x]] + E\left[(E[x])^2\right]$

$= E[x^2] - 2(E[x])^2 + (E[x])^2$

$= E[x^2] - (E[x])^2$

so $\text{Var}(x) = E[x^2] - (E[x])^2$

Q_1: $Var(u^T X) = u^T \Sigma_1 u$

$\Sigma_1 = Cov(X) = E[(X-\mu)(X-\mu)^T]$

$Var(u^T X) = E[(u^T X - E(u^T X))^2]$

$E(u^T X) = u^T E[X] = u^T \mu$

$Var(u^T X) = E[(u^T X - (u^T \mu))^2]$

$= E[(u^T (x-\mu))^2]$

$(u^T (x-\mu))^2 = (u^T (x-\mu))(u^T (x-\mu))^T$

$(u^T (x-\mu))^T = u(x-\mu)^T$

$Var(u^T X) = E[u^T (x-\mu)(x-\mu)^T u]$

$= u^T E[(x-\mu)(x-\mu)^T] u = u^T \Sigma_1 u$

↑ covariance matrix Σ

Q_3: mean vector

$\bar{x} = \frac{1}{N} \sum_{i=1}^{N} x^{(n)}$

mean subtraction

$X_s = x^{(n)} - \bar{x}$

covariance matrix

$\frac{1}{N-1} \sum_{n=1}^{N} x_s^{(n)} (x_s^{(n)})^T = \frac{1}{N-1} X_s^T X_s$

$Var(u^T X) = u^T S u$

max $u^T S u$ subject to $u^T u = 1$

Lagrange

$L(u, \lambda) = u^T S u - \lambda (u^T u - 1)$

$\nabla_u L(u, \lambda) = 2Su - 2\lambda u = 0, \; Su = \lambda u$

$Var(u^T X) = u^T S u = u^T (\lambda u) = \lambda u^T u = \lambda$

To obtain top directions max variance

select eigen vectors $u_1, u_2, \ldots$

corresponding to largest eigenvalues

$\lambda_1 \geq \lambda_2 \geq \lambda_3 \cdots$

Q_4: $\lambda_1 = 3.46$, $\lambda_2 = 0.41$

variance: $\lambda_1 + \lambda_2 = 3.87$

$PC_1 = \frac{3.46}{3.87} = 89.4\%$

$PC_2 = \frac{0.41}{3.87} = 10.6\%$

first principal component capturing 89.4% of variability in data set

second principal component captures remaining 10.6% of data set

Significance of Dimension Reduction

in higher dimension data sets, high feature counts increase complexity and memory, prone to overfitting, require much more training data

PCA turns it to lower dimension spanned by top principal components, preserve max variance, filter out dimensions with smaller eigenvalues

Trade-Off

reducing dimensions simplifies models and accelerates computation but results in loss of information if even only slightly

thus you need to determine the optimal threshold between information loss and efficiency/accuracy